# Report on "Matrix Theory Reloaded: A BPS Road to Holography"

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**Summary.** The paper treats the decoupling limit that gives BFSS matrix theory as a BPS limit of the *whole* type II theory, not just of the D-brane worldvolume. The authors call the resulting corners matrix $p$-brane theory (M$p$T). These corners have a non-Lorentzian target space with a codimension-$(p+1)$ foliation. The paper then shows four things. (i) Applied to a curved D$p$-brane background, the same limit at asymptotic infinity induces the bulk near-horizon limit, so it reproduces AdS/CFT and its non-conformal D$p$ analogues. (ii) Applying two such limits in a row gives new holographic pairs with non-Lorentzian bulks. (iii) After lifting to M-theory, each limit is a DLCQ. This motivates the conjecture that holography is a DLCQ$^n$/DLCQ$^m$ correspondence ($m > n$); for example, AdS$_5$/CFT$_4$ is DLCQ$^0$/DLCQ$^1$, and D1-D5 is an $m > n+1$ case. (iv) Undoing the limit is a U-dual, $p$-brane version of the $T\bar{T}$ deformation. It generates the curved bulk from the flat non-Lorentzian asymptotics, and it gives explicit flow equations for the abelian bosonic DBI action. A notably simple three-dimensional flow is shared by the D2- and M2-brane.

## 1. Principal achievement

The main achievement is conceptual unification. The paper links matrix theory, the Maldacena near-horizon limit, non-Lorentzian (non-relativistic string) geometry and $T\bar{T}$ deformations through one organising principle: BPS decoupling limits, where tension and RR charge cancel. The central technical result is the observation that the asymptotic M$p$T limit is the same as the bulk near-horizon limit. It turns the familiar "zoom in on the throat" step into a statement about the dual field theory. The asymptotically flat non-Lorentzian region is where the matrix gauge theory lives, and the bulk geometry follows from it. This answers a real puzzle: why the target space of matrix theory is non-Lorentzian, and what that has to do with AdS. It also gives a systematic, top-down way to generate new (including non-Lorentzian) holographic dualities, where most of the literature works bottom-up. A further concrete payoff is the family of $p$-brane $T\bar{T}$-like flow equations. They are new and can be checked at the field-theory level.

## 2. Principal drawback

The main weakness is that the strongest claims, especially the DLCQ$^n$/DLCQ$^m$ conjecture and the non-Lorentzian holographic examples, rest mostly on kinematics and analogy. There is little dynamical evidence for them.

- The new non-Lorentzian bulk geometries come from rescaling the harmonic-function length scale $\ell$ (in effect $N$) with $\omega$ by hand. The authors themselves admit they have not checked that these geometries solve the equations of motion and torsional constraints of an M$p$T supergravity, which has not yet been formulated. They have also not determined the geometries' supersymmetries, charges or isometry groups.

- No holographic observable is matched across any of the proposed new dualities: no spectrum, correlator, thermodynamics or protected quantity. So the conjecture is a reinterpretation of known cases (D3, M2, M5, D1-D5) plus a structural extrapolation, not a tested correspondence.
- The analysis is restricted to the Poincaré patch and to regimes ($p \leq 3$, small 't Hooft coupling at infinity) where other effects can enter. It is not clear how far the non-Lorentzian description at infinity is physical rather than coordinate-dependent.
- The $T\bar{T}$ flows are derived only for abelian, bosonic actions. The step that matters for matrix theory, deforming non-abelian SYM into non-abelian DBI, is still conjecture.

The paper is also very long, about 4000 lines of source. That is partly because it re-derives a lot of review material, and it makes the new results harder to pick out.

## 3. Most worthwhile follow-up

**Proposal: study the three-dimensional $p$-brane $T\bar{T}$ flow (the shared D2/M2 flow equation) as a well-defined field-theory deformation, and test its solvability.**

*Rationale.* This idea has the best ratio of impact to feasibility. The flow equation is explicit and simple, and it needs none of the unfinished M$p$T supergravity machinery. Concrete steps:

1. Apply the flow to free scalars and Maxwell theory in $2+1$ dimensions and check that the endpoints are the M2 Nambu–Goto and D2 DBI actions. Classify the operator: is it built only from $T_{\mu\nu}$ invariants, as in $T\bar{T}$ and its root-$T\bar{T}$ cousins?
2. Ask whether any of the special properties of 2d $T\bar{T}$ survive. Examples are factorisation of the deforming operator, an exact deformed spectrum on a torus or the energy of a box (Burgers-type equation), and dressing or dynamical-coordinate interpretations. This can be done first classically, with the flow on the torus energy, and then at large $N$.
3. Apply the flow to supersymmetric and non-abelian theories, starting with $\mathcal{N} = 8$ SYM in 3d and BLG/ABJM. Compare the result with the known (partial) non-abelian DBI/Myers action, and with holography via the D2/M2 near-horizon geometry. In the bulk, the paper says, the flow should interpolate from $\mathrm{AdS}_4 \times S^7$ to the full M2 solution.

*Impact.* A positive result would give the first solvable irrelevant deformation in more than two dimensions with a clear string/M-theory origin. It would also give a field-theory route to the M2 analogue of DBI, and a dynamical check of the paper's claim that the bulk is generated by a $T\bar{T}$-like flow. A negative result, meaning the flow is not solvable or not well-defined beyond the classical level, would still bound how literally the "holography as undoing BPS limits" picture can be taken. A natural second project, more ambitious but less certain, is to build M$p$T supergravity and use it to check the non-Lorentzian bulk solutions. That would directly fix the paper's main drawback.